

TASK 2 – SENSOR BASED SIMULATION

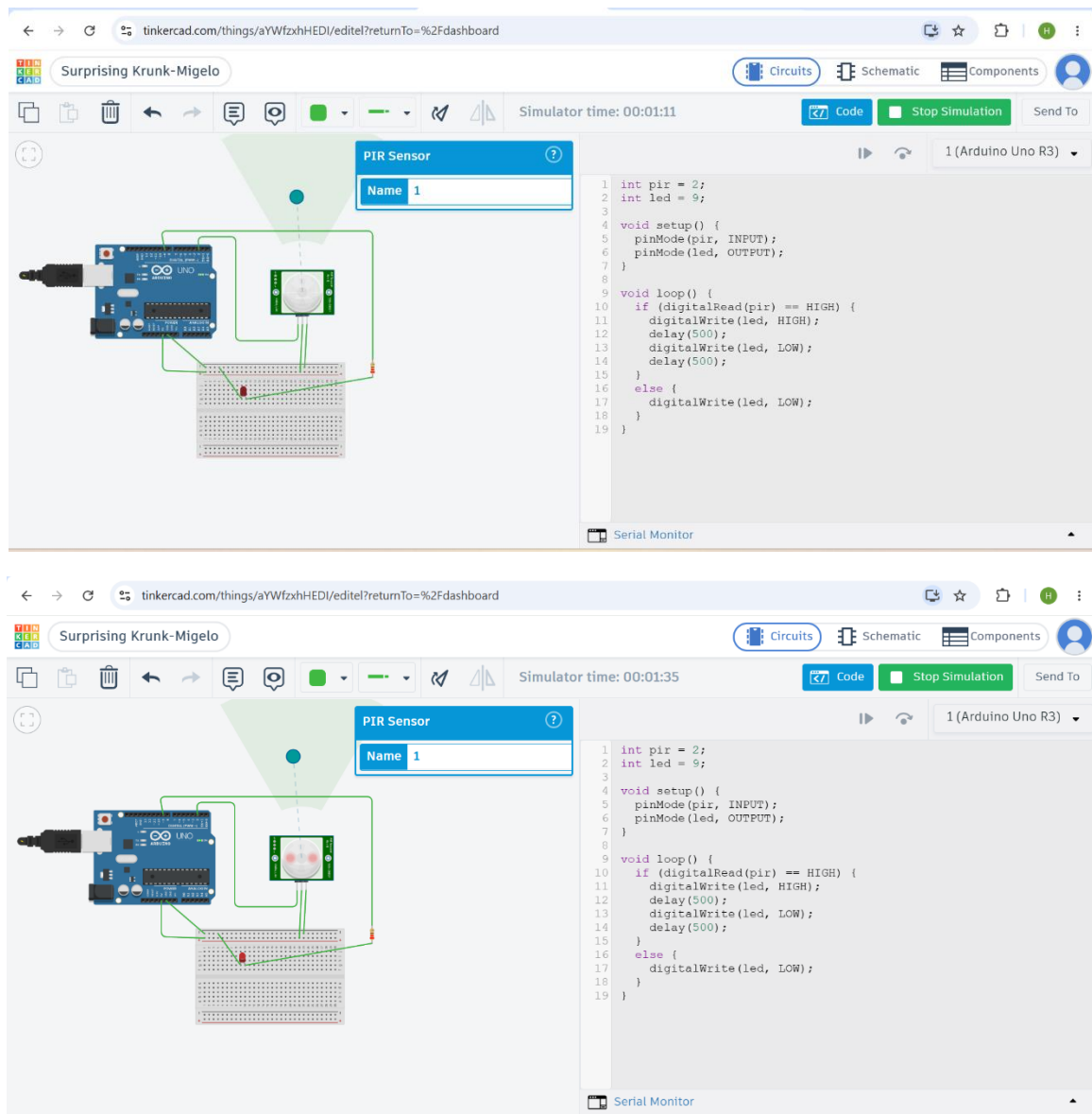


Fig : Simulation

Code Explanation

This Arduino program uses a PIR sensor to detect human movement and control an LED.

Pin Configuration

```
int pir = 2;
```

```
int led = 9;
```

The PIR sensor is connected to pin 2, while the LED is connected to pin 9.

Initialization

```
void setup() {  
  pinMode(pir, INPUT);  
  pinMode(led, OUTPUT);  
}
```

The PIR sensor is configured to receive signals, and the LED is configured to send an output signal.

Motion Detection

```
if (digitalRead(pir) == HIGH) {
```

The Arduino continuously reads the PIR sensor. When the sensor detects movement, its output becomes HIGH.

LED Operation

```
digitalWrite(led, HIGH);  
delay(500);  
digitalWrite(led, LOW);  
delay(500);
```

After detecting motion, the Arduino switches the LED ON and OFF with a 500 ms delay, causing the LED to blink.

No Motion Condition

```
else {  
  digitalWrite(led, LOW);  
}
```

When the PIR sensor does not detect movement, the Arduino keeps the LED switched OFF.

Working Principle

The PIR sensor senses movement and sends a signal to the Arduino. The Arduino processes this signal and makes the LED blink when motion is detected.

Conclusion

The program provides a simple and effective way to indicate motion using a PIR sensor and LED. It demonstrates the basic interaction between a sensor and an output device using Arduino.